

Antenatal Care ANC

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FIBOG

ANC Messages to the Students

- What is ANC?
- When we start to do it?
- Who is going to do it and to whom we concentrate?
- What diseases we are focusing on and what drugs?
- What is the role of dating and ultrasound scan?
- Common diseases we focus on?

Definition of ANC

Comprehensive health supervision of a pregnant woman before delivery

Or it is planned examination, observation and guidance given to the pregnant woman from preconception till the time of labor.

Aims of Routine ANC

- 1) Maternal health check.
- 2) Evaluation of fetal health & development.
- 3) Screening for diseases
- 4) Analysis of risk of complication.
- 5) Advice & education (childbirth , childcare, nursing)

We should focus on

- * Communications is a key factor between mother & health professional + between Different health professionals providing the service.
- * Mother is the center of all cares and screening done along the process of pregnancy follow up
- * ANC models changed on last decade / care outside Hospital / more care by MW

We can divide Explanation and Advice to be about:

- 1) Minor complaints : To know about symptoms, including nausea, heartburn, constipation, shortness of breath, dizziness, swelling, backache, abdominal discomfort and headaches.
- 2) Major complaints : For complicated previous or suspected conception eg. Blood incompatibility, eclampsia

Classification of ANC

Consultant care :Women with complex pregnancies that the vast majority of their antenatal care is provided by a hospital-based obstetric team

Booked under the Midwives :Women have pregnancies where there are no overtly complicating factors and these women usually have community-based care

Share care :Combination of both in Caesarian section cases

How frequent to be done?

According to the mother condition:

- * Nulliparous uncomplicated = 10 visits
- * Parous uncomplicated = 7 visits
- * More frequent / in high risk pregnant women

The important visits of ANC

- Preconception visit
- Booking visit from 10th weeks on
- Second trimester visits
- 36-38 weeks
- 41-42 weeks

For those who planning to be pregnant

- Detailed education of conception, delivery methods, maternity and breast feeding
- Screening of diseases
- Vaccinations
- Drugs to be taken
- Habits to be stopped eg. Smoking, Alcohol, drugs
- Family sharing especially the husband
- Explaining the importance of ANC

Booking visit : includes;

- **Pregnancy confirmation** : Symptoms, Blood, Urinary, U/S (Heart Beat)
- **Pregnancy dating** : By Expected Date of Delivery using last menstrual period: (if menses was regular) by adding 9M & 7D.
- **Early U/S** : Week 10 -14, Crown Rump Length (CRL), Head Circumference (HC), Down screening, Multiple pregnancies (Chorionicity), detect up to 80% of lethal abnormalities

Booking History & Examination

History:

- Age and racial
- Past medical, HTN, DM
- Past Obstetric and gynaecological,
- miscarriage, C/S, etc.
- Family and social history, eg. Hemoglobinopathy, smoker, alcohol

Examination

- Weight & Height, BMI
- Accurate measurement of blood pressure.
- Abdominal examination, uterine size, scar
- Systemic examination

The Booking Investigations

- Blood tests: Full Blood Count , Blood group and RBC antibodies , Hemoglobin studies
- Urine analysis
- Screening for Infection, Rubella, Syphilis, HIV, HBV
- Screening for fetal abnormalities

Second Trimester visits

- Assessment of maternal health & fetal growth & wellbeing are persuaded through these visits.
- The results of tests performed at 1st trimester visit are reviewed with the mother
- The results of the U/S scan for fetal abnormality are also reviewed.
- Any incidental maternal symptoms are dealt with eg. Screening for Diabetes, Hypertension
- This period is also important in insuring the education of the woman regarding the rest of pregnancy & her delivery.

36-38 Weeks

- Preparation for labour
- Recognition of active labour
- Anticipation of any problems regarding the prospective delivery.
- Symphysio fundal height ,fetal lie, presentation
- Vaginal examination; Checking pelvis and engagement of fetal presenting part
- Debate about Mode of delivery , Breastfeeding , planned contraception

41-42 Weeks

- With accurate pregnancy dating, true post dates pregnancy are identified.
- At this visit , a Committee decision is taken as to whether an induction of labour is appropriate, this is current practice because of the reported association between post dates pregnancy & pregnancy outcome.

- Induction of labour usually recommended by completed 41 week & there are two main methods:
 1. Amniotomy
 2. Medical methods using Prostaglandin or Oxytocin

Antenatal care in certain Medical conditions:

ANC in Gestational Diabetes:

Risk assessment of gestational diabetes at the booking appointment:

- BMI above 30 kg/m²
- previous macrosomic baby weighing 4.5 kg or above
- previous gestational diabetes
- family history of diabetes (first-degree relative with diabetes)
- minority ethnic family origin with a high prevalence of diabetes.

N.B.: Offer women with any one of these risk factors testing for gestational diabetes

ANC monitoring in gestational diabetes include:

- Monitoring blood glucose: Advise pregnant women to test their fasting, pre-meal, 1-hour post-meal and bedtime blood glucose levels daily during pregnancy.
- Target blood glucose levels: Fasting: 95 mg/dl, 1 hour after meals: 140 mg/dl, 2 hours after meals: 115 mg/dl, Diabetes on insulin or glibenclamide : >72 mg/dl
- Monitoring HbA1c: Measure HbA1c levels in all pregnant women with pre-existing diabetes at the booking appointment to determine the level of risk for the pregnancy; and also in the second and third trimesters to assess the level of risk for the pregnancy.
This risk for women with pre-existing diabetes increases with an HbA1c level above 48 mmol/mol (**6.5%**). However; we should not use HbA1c levels routinely to assess a woman's blood glucose control in the second and third trimesters of pregnancy.
- Monitoring fetal growth and wellbeing: Ultrasound monitoring of fetal growth and amniotic fluid volume every 4 weeks from 28 to 36 weeks. If risk of fetal growth restriction present after 38 weeks, offer fetal umbilical artery Doppler recording, fetal heart rate recording and biophysical profile testing

ANC in Hypertension in Pregnancy

- Pre-pregnancy advice for women with treated hypertension; include AHD taking in pregnancy and the risk of congenital abnormalities related to some of the drugs
- Women with hypertension in pregnancy have a blood pressure target set below 150/100 mmHg or, if they also have target organ damage like Left ventricular hypertrophy, chronic kidney disease and hypertensive retinopathy, below 140/90 mmHg.
- Antihypertensive treatment should aim to lower blood pressure from the moderate or severe range, while avoiding excessive reductions that may affect fetal growth.

- Pre eclampsia risk factor assessment:



High risk factors:

- *Hypertensive disease in a previous pregnancy
- *Chronic kidney disease
- *Autoimmune disease, such as SLE or APL syndrome
- *Type 1 or type 2 diabetes
- *Chronic hypertension.

Moderate risk factors:

- *First pregnancy
- *Age 40 years or older
- *Pregnancy interval of more than 10 years
- *BMI of 35 kg/m² or more at first visit
- *Family history of pre-eclampsia
- *Multiple pregnancy.

In the presence of pre-eclampsia risk at the booking appointment, we offered a prescription of 75 mg of aspirin to take daily from 12 weeks until birth.

Lifestyle considerations in ANC for Pregnant women:

Adequate nutrition

* Calories (2500 Kcal/day)

* Protein (60gm/day)

* Calcium (1.2 gm /day)

Iron (30 mg of ferrous / day)

- *Animal source:* red meat
- *Plant source :* dark green vegetable
- *Drug sources :* ferrous gluconate, ferrous fumarate, ferrous sulfate
- A dose of 30mg elemental iron per day should be enough for most woman
- 60-100 mg/day is given for large women, twin, and those women who book for ANC late in pregnancy
- Anemic woman should take Iron 200 mg/day
- Iron requirements are increased only in the later half of the pregnancy
- Insufficient iron in the diet leads to maternal iron deficiency anemia.

*** Note: if iron and calcium are prescribed, they should be taken 6-12 hours apart.

Sexual activity

- Sexual activity is allowed in moderation
- It's to be avoided in pregnant women with threatened miscarriage, preterm labor or APH

Travel

- Travel is allowed when comfortable
- Car safety belts have to be adjusted to be comfortable for woman
- Those traveling more than three hours (either by car or airplane) must take a break every 4 hours and walk for about of minutes to decrease the risk of DVT

Weight gain (11-16 kg)

- Normal weight women should gain 11.5-15 kg
- Underweight women should gain 12.5-18 kg
- Obese women should gain no more than 7 kg

Advice for the mother

Alarming Symptoms and signs

- Vaginal Bleeding
- Severe edema
- Escape of fluid from the vagina
- Abnormal gain or loss of weight
- Decrease or cessation of fetal movement
- Severe headache
- Epigastric pain
- Blurred vision
- Fever
- Abdominal pain

Fetal wellbeing

Assessment of fetal well being in a low risk pregnancy

- Symphysis–fundal height should be measured and recorded at each antenatal appointment from 24 weeks.
- Fetal kick count: at least 10 movements/12 hours (a change in the kick count is more important than the absolute number); not routinely offered
- Fetal movements: absence proceeds intrauterine fetal death (IUFD) by 48 hours
- Fetal heart sounds (bradycardia and or tachycardia indicate possible fetal compromise) ; but routine auscultation is not necessary
- Fetal presentation should be assessed by abdominal palpation at 36 weeks or later, when presentation is likely to influence the plans for the birth.
- Suspected fetal malpresentation should be confirmed by an ultrasound assessment.

Conclusion

- Antenatal care is an essential aspect of health care delivery for improving pregnancy outcome.
- By this service we can detect high risk pregnancies and we can direct them for proper management

Good Luck